

Chapter12 Theory of Computation

Functions and Their Computation

- **Function:** A mapping from input values (Domain) to output values (Range).
- **Computable Function:** A function for which an **algorithm** exists that can determine the output for any given input. (Also called **Turing computable**).
- **The Limit:** There are more functions than there are programs. Therefore, **Non-computable functions must exist**.

Turing Machine(圖靈機)

- **Components**
 - **Tapes:** Cells preserve data(Input and output).
 - **Head:** Read, update each cell.
 - **State Controller:** Condition of machine at anytime.
- **Church-Turing Thesis: Algorithms = Turing Machine**
 - Any function that is computable by an algorithm is computable by a Turing Machine.

Universal Programming Language

- **A language with which a solution to any computable function can be expressed.**
- **Bare Bones:** Most simple language.
 - **Commands**
 - **Variable:** Can store numbers.
 - **Clear:** Sets the variable to zero.
 - **Incr:** Increments the variable by 1.
 - **Decr:** Decrements the variable by 1.
 - **While...End:** Repeats execution (loop) as long as the variable is not zero.

The Halting Problem

- **Given the encoded version of any program, return 1 if the program is self-terminating, or 0 if the program is not.**

```
def H(P):
    if P 執行自己會停機:
        return 1
    else:
        return 0
def Strange(P):
    result = H(P) #
    if result == 1:
        while True:
            pass
    else:
        return
```

Complexity Of Problems

- **Time Complexity:** The number of instruction executions required.
- **NP v.s. P**
 - **Class P:** All problems in any class $O(f(n))$, where $f(n)$ is a polynomial.
 - **Class NP:** All problems that can be solved by a nondeterministic algorithm in polynomial time.
 - **Nondeterministic algorithm:** An “algorithm” whose steps may not be uniquely and completely determined by the process state.
 - **NP = P?**
 - Whether the class NP is bigger than class P is currently unknown.
 - Maybe all NP problems have a solution that is very fast.

Public-Key Cryptography

- **Key:** A value used to encrypt or decrypt a message
 - **Public key:** Used to encrypt messages

- **Private key:** Used to decrypt messages
- **RSA:** A popular public key cryptographic algorithm.
 - Give two prime numbers x and y , it is easy to get $N = x * y$
 - But it is hard to find x and y , when only give N (maybe it is impossible)